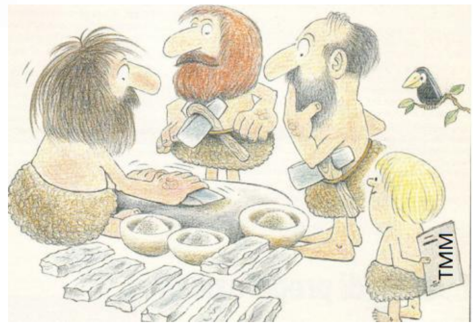
Politecnico di Torino

TECHNOLOGY of METALLIC MATERIALS

Mechanical & Automotive Engineering
A.Y. 2025-2026



Lecture 2.
METALS FUNDAMENTALS – SECOND PART

Giovanni Maizza (giovanni.maizza@polito.it)

1

Mechanical Properties of Metals

- **Classification**
- **Electronegativity (difference)**
- **Slip in Crystal structures** (BCC, FCC, EXC)
- **Crystal anisotropy** (elastic) and microstructure anisotropy
- **Polymorphism** (change of crystal lattice with temperature)
- Manufacturing introduces different types of nc/micro/macro and concentration of defects
- Metals and alloys fabrication
  - General casting
    - Sand casting
  - Integrity of castings
  - Casted alloys and Market share
- Various types of manufacturing processes
- Metallic flow in atomotive industry
- Sustainable manufacturing
- Control of microstructure (micro and nc) by manufacturing
- **Lattice defects** (surface, vacancies, dislocations, GBs, slip of dislocations, slip systems in single crystal)
- Self assessment questions

2

Why Metals and Alloys are unique?

They are structural materials used to fabricate functional- and load bearing components

What is the difference between metals and alloys vs. other engineering material families:
- polymers
- ceramics
- composites

What are the outstanding, features, properties and attributes of metals and alloys for engineering applications:

- ***Many methods of part shaping (wrought, casting, PM, joining, etc.)***
- ***Many methods of strengthening (large variety of mechanical properties***)
- ***Excellent compromize between high-strength and high-toughness***
- ***Very sensitive to cooling/heating rates (fine microstructure)***
- ***Can combine functional and structural performance***
- ***Can undergo significant plastic deformation***

Inconveniences: affected by environmental degradation, on-service degradation and embrittlement, very sensitive to strain rate, sensitive to transition temperature

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Classification of Metals

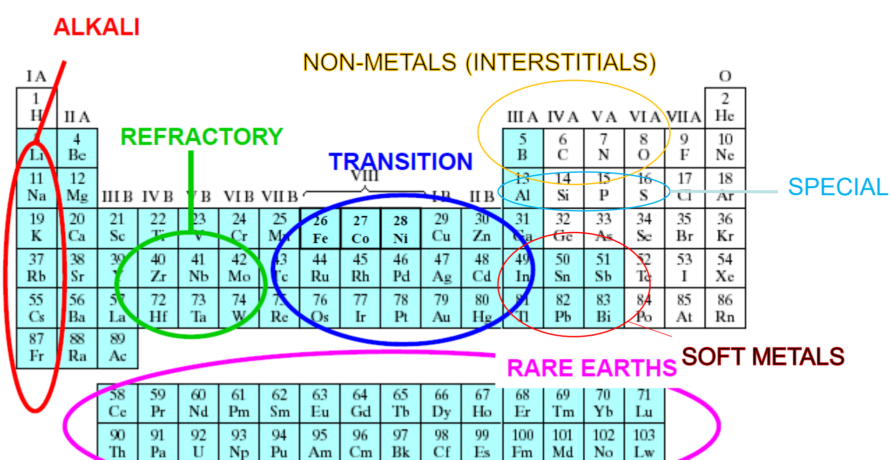

Figure 1-3 Periodic table of the elements with those elements that are inherently metallic in nature in color.

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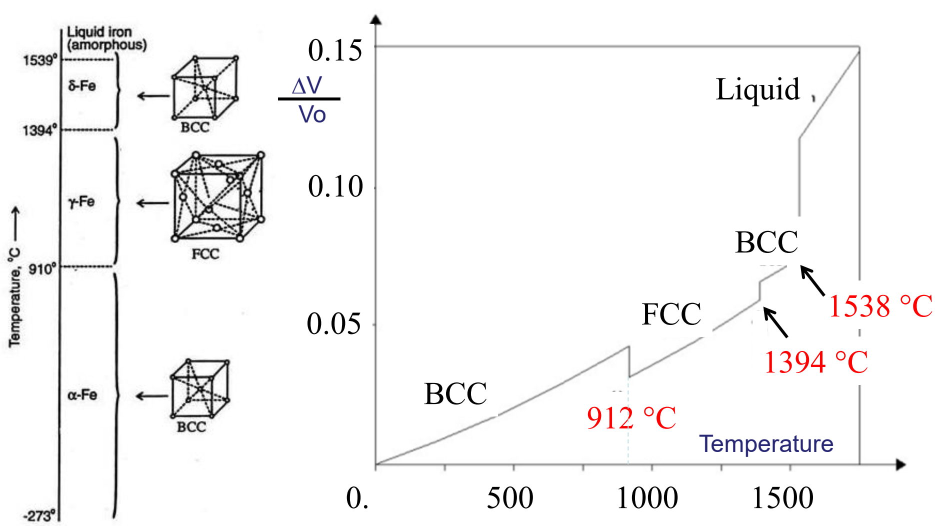
ELETTRONEGATIVITY

	Group																	
Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H 2.1																	He 0
2	Li 0.98	Be 1.57											B 2.04	C 2.55	N 3.04	O 3.44	F 3.98	Ne 0
3	Na 0.93	Mg 1.31											Al 1.61	Si 1.9	P 2.19	S 2.58	Cl 3.16	Ar 0
4	K 0.82	Ca 1	Sc 1.36	Ti 1.54	V 1.63	Cr 1.66	Mn 1.55	Fe 1.83	Co 1.88	Ni 1.91	Cu 1.9	Zn 1.65	Ga 1.81	Ge 2.01	As 2.18	Se 2.55	Br 2.96	Kr 0
5	Rb 0.82	Sr 0.95	Y 1.22	Zr 1.33	Nb 1.6	Mo 2.16	Tc 1.9	Ru 2.2	Rh 2.28	Pd 2.2	Ag 1.93	Cd 1.69	In 1.78	Sn 1.96	Sb 2.05	Te 2.1	I 2.66	Xe 2.6
6	Cs 0.79	Ba 0.89	La 1.1	Hf 1.3	Ta 1.5	W 2.36	Re 1.9	Os 2.2	Ir 2.2	Pt 2.28	Au 2.54	Hg 2	Tl 2.04	Pb 2.33	Bi 2.02	Po 2	At 2.2	Rn 0

White= No data	0-.66	.66-1	1-1.33	1.33-1.66	1.66-2	2-2.33	2.33-2.66	2.66-

Electropositive elements bond with other elements by releasing their electrons and are particularly reactive (Metals). Also electronegative elements are very reactive.

Allotropy, Polymorphism: Fe



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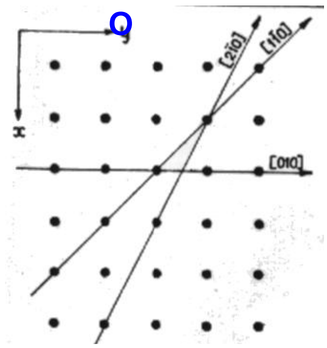
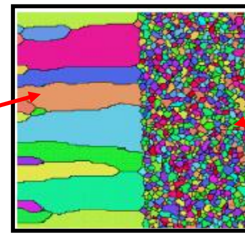
Anisotropia: Cristallina e Microstrutturale

CRISTALLO



MICROSTRUTTURA DA GETTO

Grani colonnari (anisotropi)



Grani Equiassici (isotropi)

	Modulus of Elasticity (GPa)		
Metal	*[100]*	*[110]*	*[111]*
Aluminum	63.7	72.6	76.1
Copper	66.7	130.3	191.1
Iron	125.0	210.5	272.7
Tungsten	384.6	384.6	384.6

Bartocchi, p.70

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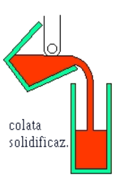
Fabrication of Metals and Alloys by microstructure control

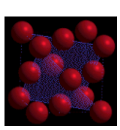
solidification

Liquid metal → Unit cell → crystal → Single crystal → Final part: turbine blade

→ Polycristals → Engineering products.

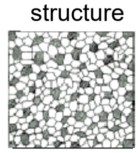


FCC, BCC, ec, allotropy



Dendritism microdefects, Lattice defects

structure



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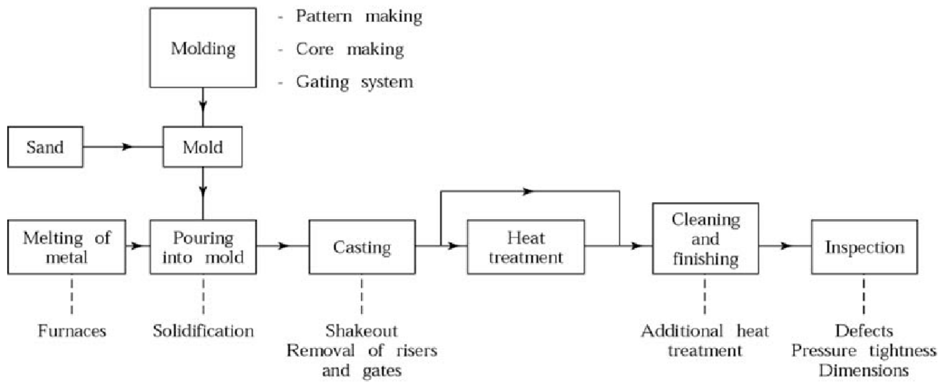
Sand casting process



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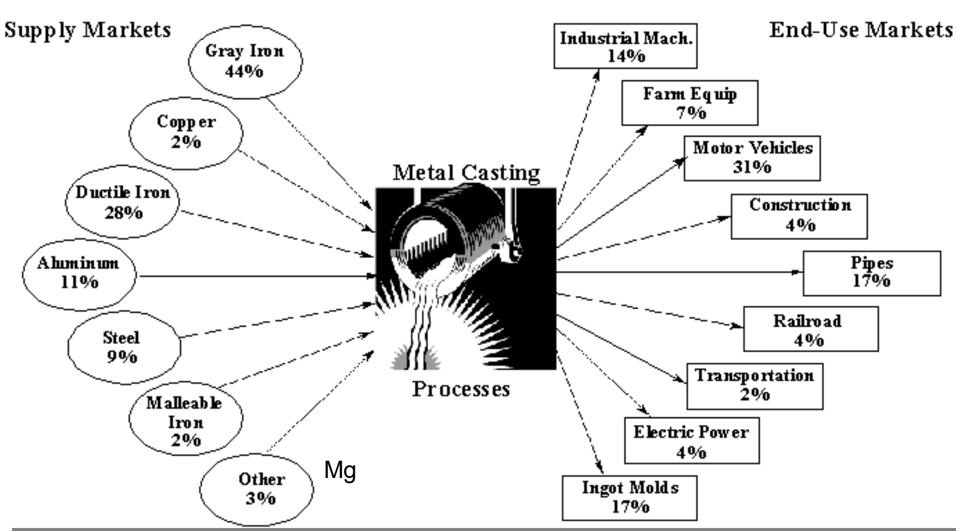
Casted Alloys - Supply/Market Share



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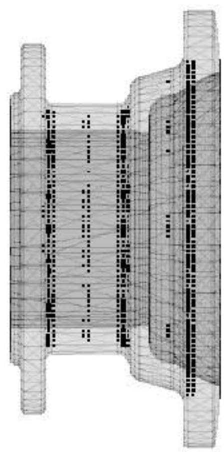
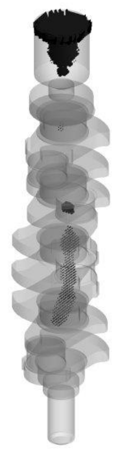
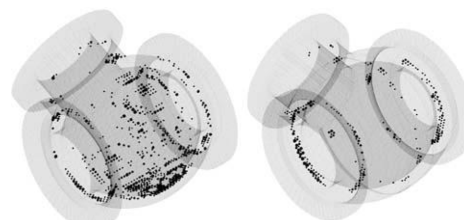
Soundness of Castings by Xray



Auto wheel



Crankshaft



Piping

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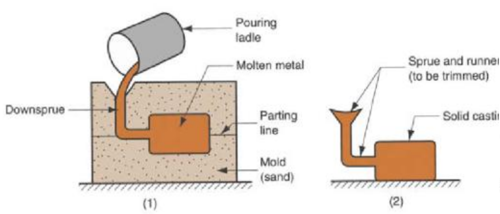
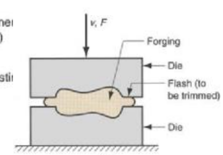
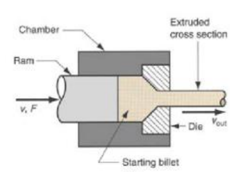
Relevant Shaping Metahods of Metals and Alloys



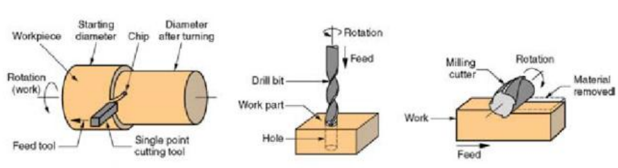
a) *casting*



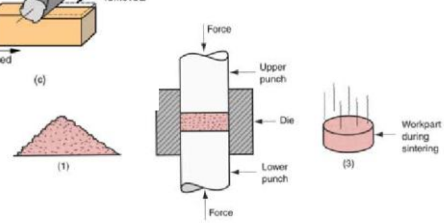
b) *Forging*



Extrusion



c) *Machining*



d) Powder metallurgy

e) - Surface heat treatment
f) - Joining and welding

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Sustainable Manufacturing

Waste and scraps from manufacturing shall be minimized
- casting and PM cause little waste
- maching is a wasteful operation

Technological tendency:
- **net shape** or **near-net shape** products require little machining to achieve the final part geometry and tolerances

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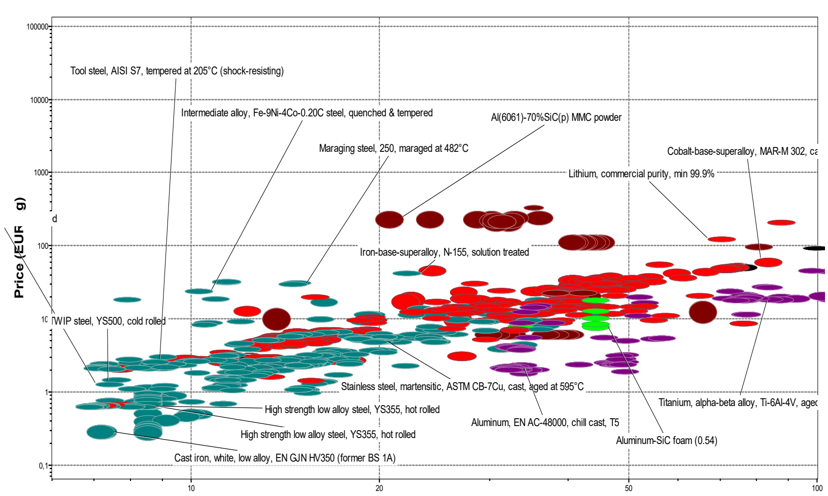
Price vs. Embodied Energy, Recycling for 6 Metals and 6 Alloys



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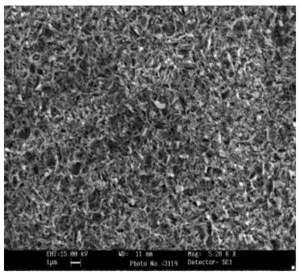
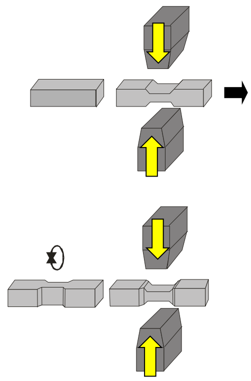
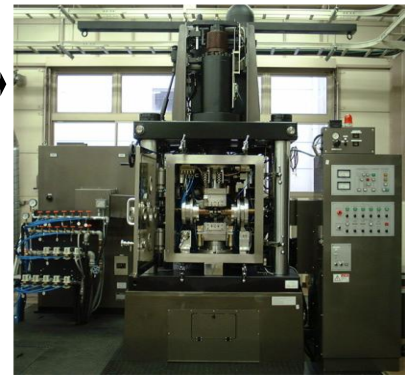
Microstructure Control: severe multidirectional plastic deformation

ultrafine alloys







Simulator: 25t hot rolling & Forging

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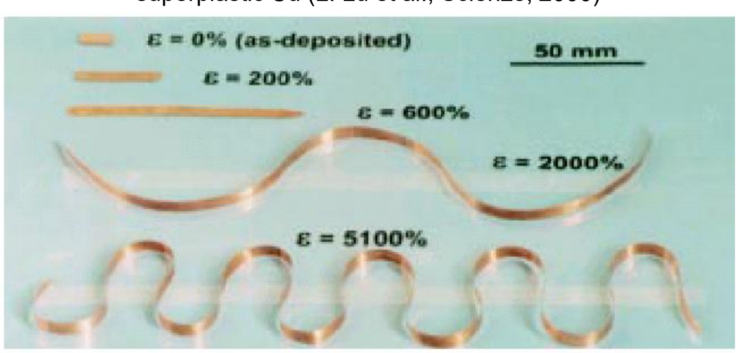
Microstructure Control: nc Metals (by electrodeposition)

superplastic Cu (L. Lu et al., Scienze, 2000)



nCu samples before and after tensile testing at Tamb and very low rates of deformation.

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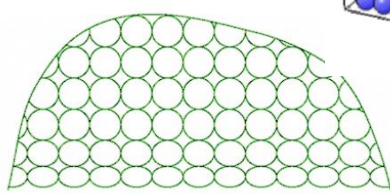
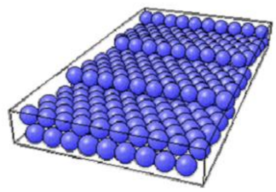
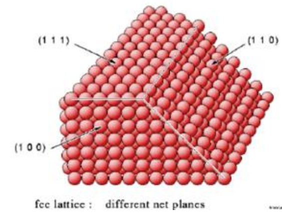
Lattice defects: surfaces

Prone to corrosion



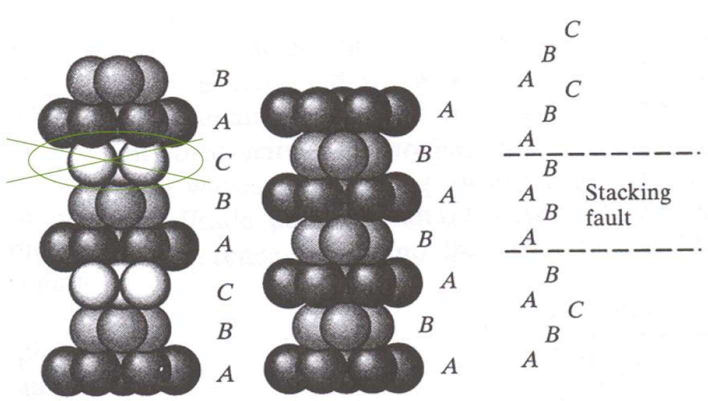




It is the bounding region between the metal and the environment $E_{free\_energy\_surface} < E_{internal}$

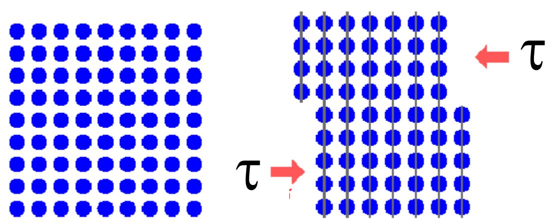
Crucial in the electrochemical degradation phenomena

Stacking Faults in FCC crystals



(a) FCC stacking (b) HCP stacking (c) Stacking fault (HCP stacking) in FCC structure

Plastic deformation (slip) of single crystals (Brief Review from STM)



Note that slip is only possible under an imposed shear stress

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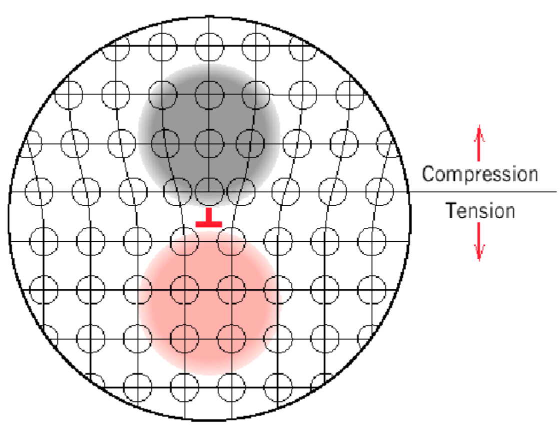
Dislocations exhibit a bipolar strain field



Dislocations in materials induce an heterogeneous strain field around the dislocation core due to the lattice distortion of the respective cores.
The distortion decreases markedly with the distance from the dislocation cores.
Edge dislocations induce in the lattice compression, tensile and shear strain

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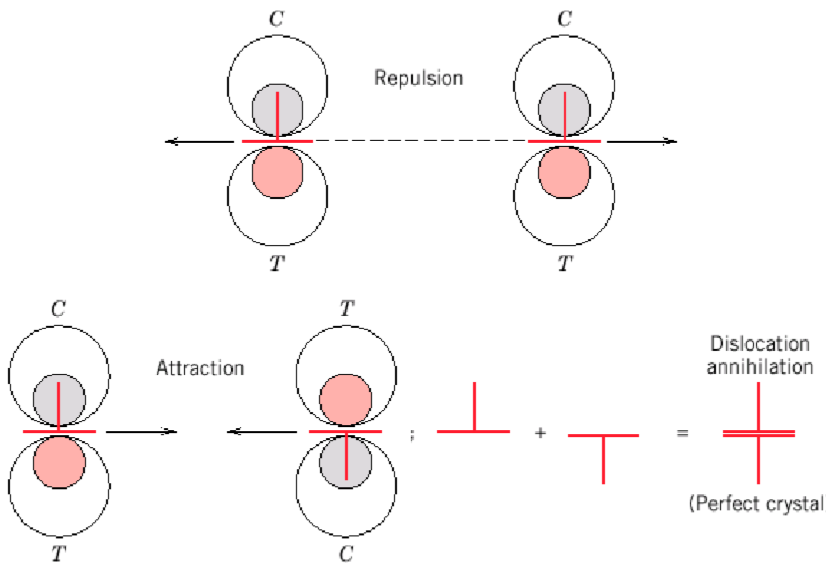
Dislocations interactions



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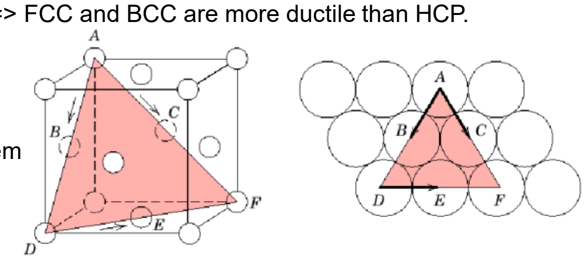
Slip systems in single crystals

Crystals exhibit special planes, called slip planes, over which dislocations move easily as compared to other planes. Such planes contains a high packing factor (atom density).
Dislocations preferentially slip over slip planes and along **highly dense directions**.
High dense plane families and high dense direction families (slip direction) identify a slip system.
BCC, FCC and EXC crystals have their own characteristic slip systems
Because the distance between atoms is smaller than average one, the interplanar distance (to the normal direction) is larger than its average counterpart allowing easier relative slip of these planes w.r.t. other planes
BCC and FCC crystal possess more slip systems than HCP crystal, thereby have more freedom to let dislocation to propagate; this confers to metals larger ductility
Consequently, => FCC and BCC are more ductile than HCP.

FCC slip system



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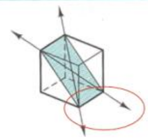
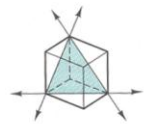
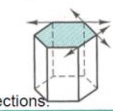
Slip systems in Metal crystals

Crystal structure	Slip plane	Slip direction	Number of slip systems	Unit cell geometry	Examples
bcc	{110}	$\langle\bar{1}11\rangle$	$6 \times 2 = 12$		α-Fe, Mo, W
	{112}	//	6 x 2 = 12		
	{123}	//	6 x 4 = 24		
			In total: 48		
fcc	{111}	$\langle 1\bar{1}0\rangle$	$4 \times 3 = 12$		Al, Cu, γ-Fe, Ni
hcp	(0001)	$\langle 11\bar{2}0\rangle$	$1 \times 3 = 3$		Cd, Mg, α-Ti, Zn

Dislocations can only move on slip planes along slip directions.
The slip planes and directions are the closest packed in a lattice

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End of Lecture n.2

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Self-assessment Questions 1

1. Performance vs. properties: which manufacturing factors affect the mechanical response of a tensile sample vs. a true structural part
2. Performance vs. properties: compare the tensile strength of a material sample vs. that extracted from a structural part?
3. Performance vs. properties: following questions n.1 and n.2, guess similar answers assuming a complex geometry and large size of the structural part
4. Why the dimensional scale of microstructure does affect the material properties?
5. Which are the instruments allowing observation of microstructures a different dimensional scales?
6. How crystal-sensitive material properties are measured or diagnosed ? and list a few of them.
7. How the individual scales of microstructure do interact to each other?
8. Which lattice defects do affect mechanical properties ? List a few of them.
9. Does cold rolling affect the isotropy of propertie? Which is most penalized?
10. Are the properties isotropic after annealing a cold rolled alloy?
11. How do strength and toughness affect the cold worked parts
12. Why polycrystalline materials do exhibit isotropic mechanical properties in spite of anisotropic properties across individual grains
13. How shall be designed the microstructure for best strength and toughness combination

Self-assessment Questions 2

11, Why of the exceptional combination of strength, toughness and sharp edge in Damascus swords: justify each performance with the received manufacturing proces
12. How shall be designed an optimal microstructure to withstand abrasive wear?
13. Performance vs. properties: following questions n.1 and n.2, guess similar answers assuming a complex geometry and large size of the structural part
14. Why strain hardening ability is maximized in Cu and Cu-alloys?
15. Guess the degree of coherency of the precipitate interface at maximum peak strength upon correct aging?
16. What is the strengthening method enabling the maximum strength and toughness combination?
17. Which strengthening method does exert uniform plastic strain?
18. Assuming the presence of manufacturing defects in industrial parts, which mechanical property shall priorily be used to meet safe design requirement?
19. Does the base metals in alloys affect their strength ?
20. What is the most effective dislocation source in metals?
21. Which are the primary factors in precipitation hardening?
22. Why substitutional atoms determine an incresed strength in alloys?
23. Why of the upper and lower yiels stress in some alloys?
24. Why of the Luders’ bands during tensile test of some alloys

Self-assessment Questions 3

11. Shall the angle of GB be small (<15°) or large (>15°) to enable strain hardening?
12. Is thermal expansion a microstructure-sensitive or insensitive property?
13. Why the first stage of aging is accompanied by a decrease in strenght?
14. What is the most important performance which must be exhibited by castings in practical applications and how it it is met in the manufacturing practice?
15. What is the simplest and most economical method to strengthen a metal?
16. In precipitation strengthening shall precipitates form inside the grains or at their GBs?
17. Explain the phenomenon of Cottrell atmospheres and its consequences?
18. What is the common factor shared by all strengthening methods except that for the fiber strengthening method?
19. Explain why toughening of castings is achieved using the eutectic modification method?
20.What is the main strengthening method in HSLA steels?
21.What is the main strengthening method of aluminum alloy?
22. State the Schmid law?
23. Explain how the yield stress of an alloy changes upon cyclic strain hardening
24. How does the SFE affect the strain hardening of alloys
25. Which among BCC and FCC is more ductile? Explain the reason.

Self-assessment Questions 4

26. Explain why strengthening by solid solution is proportional to √%C
27. Explain how ductility varies after solid solution strengthening of an alloys
28. Explain why certain alloy elements in certain alloys cause strengthening while other elements cause softening
29. Explain which lattice defects, other than dislocations, contribute to strain hardening and why
30. Clarify the importance of Orowan’s equation on work hardening
31. Why are dislocation sources are essential for strain hardening
32. Clear up in which conditions plastic deformation occurs by twinning rather than by slip
33. Explain the micro- and macroscopic phenomena of strain hardening with reference to both Ludwik-Hollomon and Taylor laws
34. Justify the large strain hardening coefficient of Co-alloys and Cu-alloys vs. that of cold rolled steels
35. Elucidate the role of SFE on dislocation arrangement upon ultrafast plastic deformation
36. Compare the consequences of strain aging and dynamic strain aging
37. Detail why strengthening by grain refinement enhance both strength and fatigue resist.

End of Lecture n.2

Leonardo quote

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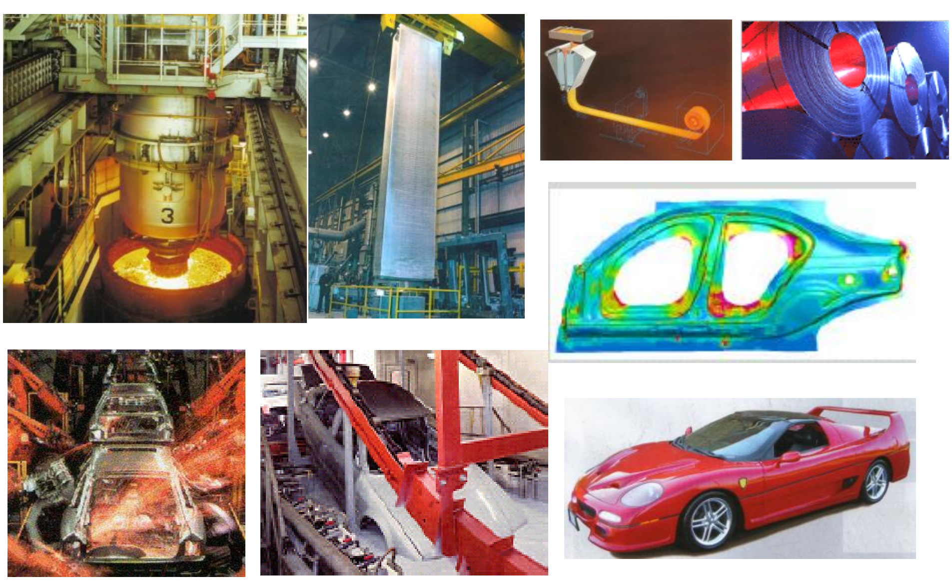
Manufacturing of Metallic Materials



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How to Detect Materials Microstructure)

Transmission Electron Microsc. (TEM)
Very high magnification (> 1 nm)

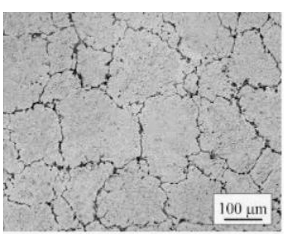
1250 keV U of Antwerp (Belgium)

Scanning Electron M.
(SEM) (>50 nm)

Optical M. (OM)
(>1 μm)

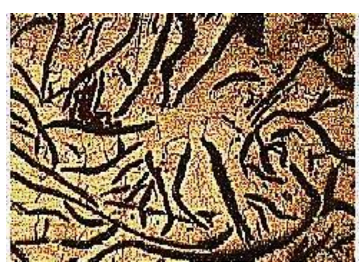
Xray diffractometer
(>0.1 nm)

Typical Microstructures of Engineering alloys

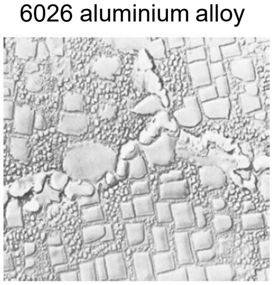


6026 aluminium alloy

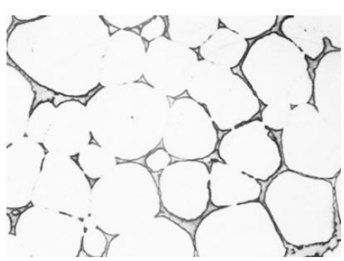
cast iron ‘bull eye’



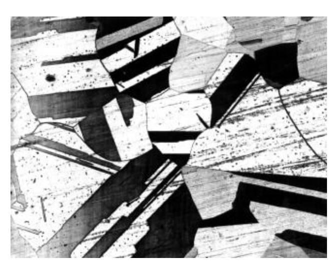
Lamellar cast iron



Ni-based superalloy

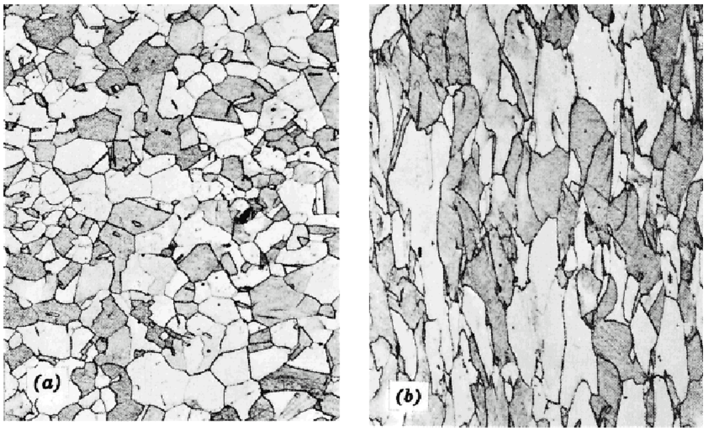


Mg alloy casting



Brass (Cu-Zn)

CW in Polycrystalline Metals



before after

A larger plastic deformation, causes a larger elongation of the grains along the direction of applied stress

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Materials Substitution Table

Material	Con-sumption	Value	Import content	Energy	Resource life	Security of supply	Ease of use	Recycle potential
Steel	1	0	4	2	4	4	4	3
Aluminium	2	3	3	0	3	3	4	4
Copper	2	2	3	2	3	2	5	4
Lead	3	3	3	3	2	3	5	5
Zinc	3	3	2	1	2	3	4	3
Tin	4	4	2	3	1	1	–	2
Nickel	4	4	1	3	4	4	–	1
Magnesium	4	4	2	0	5	5	3	3
Mercury	5	5	0	2	1	0	–	4
Glass	2	2	5	3	5	5	2	3
Brick	0	2	5	5	5	5	2	0
Concrete	0	1	4	4	5	5	3	0
Timber	1	1	2	5	4	3	4	2
Plastic	2	1	2	1	3	4	4	1

0 = bad, 5 = excellent.

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Raw Material Cost for Stainless Steels

Material	Price $/tonne
Iron (as scrap)	149[a]
Chromium (as high carbon ferro-chromium)	924–1073[a]
Nickel (as the pure metal)	6755[b]
Molybdenum (as ferro-molybdenum)	10000[a]
Manganese (as ferro-manganese)	396[a]
Copper (as the pure metal)	2323[b]
Titanium (as ferro-titanium)	3465[a]
Niobium (as ferro-niobium)	15000[a]

Note: Costs are for alloy content, e.g., the cost given is per tonne of chromium not per tonne of ferro-chromium.

[a] Private communication from suppliers, December, 1996.
[b] London Metal Exchange, December, 1996.

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MATERIAL PROPERTIES INVOLVED IN THE SELECTION OF MATERIALS

1) Chemical composition (%)
2) Contamination of contents by corrosion products
3) Corrosion characteristics in:
   - Atmosphere
   - Water
   - Soil
   - Chemicals
   - Gases
   - Molten metals
4) Creep characteristics @ temperature range
5) Crystal structure
6) Damping coefficient
7) Density (g/cm^3)
8) Effect of cold working
9) Effect of high temperature on corrosion resistance
10) Effect on strength after exposure to:
    - Hydrogen
    - High temperatures
11) Electrical conductivity (mho/cm)
12) Electrical resistivity (Ω/cm)
13) Fire resistance
14) Hardenability
15) Maximum temperature not affecting strength (°C)
16) Melting point (°C)
17) Corrosion factor (rapidity of corrosion)
18) Susceptibility to corrosion:
    - General
    - Hydrogen damage
    - Pitting
    - Galvanic
    - Corrosion fatigue
    - Fretting
    - Stress corrosion cracking
    - Corrosion/erosion
    - Cavitation damage
    - Intergranular
    - Selective attack
    - High temperature
19) Thermal coefficient of expansion ($°C^{-1}$)
20) Thermal conductivity (W/m °C)
21) Wearing quality:
    - Inherent
    - Via heat treatment
    - Via plating

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Atomic and Ions' sizes

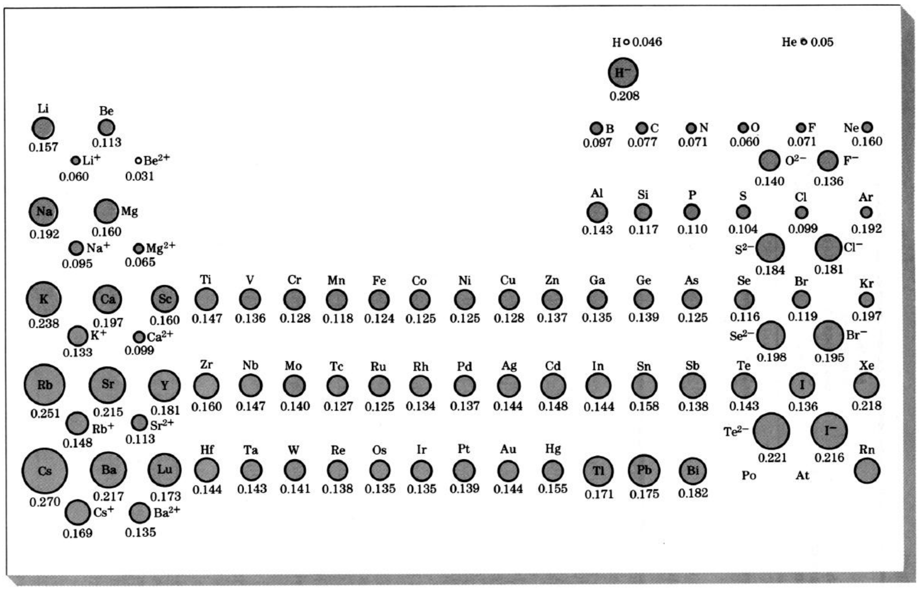

FIGURE 2.6 Relative sizes of some atoms and ions. Values are given in nanometers for the radii of the atoms and ions. Metallic radii are given for atoms where applicable. (*Adapted from F. M. Miller, "Chemistry: Structure and Dynamics," McGraw-Hill, 1984, p. 176.*)

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